MOCK TEST 20

📖 READING

5 parts · 35 questions

⏱ 60 minutes

Vaqt: 60 daqiqa. Javoblarni javoblar varaqasiga o'tkazishni unutmang.

PART 1

Questions 1–7

Read the text. Fill in each gap with ONE word. You must use a word which is somewhere in the rest of the text.

ELEPHANTS SENSE 'DANGER' CLOTHES

St Andrews University researchers discovered that elephants could recognise the degree of danger posed by various groups of individuals. The study found that African elephants always reacted with fear when they detected the smell of clothes previously worn by men of the Maasai tribe. They are known to demonstrate their courage by spearing **Q1**_____ The elephants also responded aggressively to red clothing, which defines traditional Maasai dress.

However, the elephants showed a much milder reaction to **Q2** _____ previously worn by the Kamba people, who do not hunt elephants and pose little threat.

The **Q3**_____ first presented elephants with clean, red clothing and with red clothing that had been worn for five days by either a Maasai or a **Q4**_____ man.

They revealed that Maasai-smelt clothing motivated elephants to travel significantly faster in the first minute after they moved away.

They then investigated whether elephants could also use the colour of clothing as a cue to classify a potential threat and found the elephants **Q5**_____ with aggression when they spotted red but not white cloth. This suggested that they associated the colour red with the Maasai.

The researchers believe the distinction in the elephants' emotional reaction to smell and **Q6**_____ might be explained by the amount of risk they sense. They might be able to distinguish among different human groups according to the level of risk they posed.

«We regard this experiment as just a start to investigating precisely how elephants 'see the world', and it may be that their abilities will turn out to equal or exceed those of our closer relatives, the monkeys and apes,» **Q7**_____ added.

PART 2

Questions 8–12

You work for a travel agent. On the page there are descriptions of eight holidays. Decide which holiday (letters A to H) would be the most suitable for each person described below (8–12).

8

Person 8

Kevin is a very busy investment banker who wants a week off work. He likes scenery and fresh air. He is very fit and he enjoys walking. For this holiday he wants to leave the beach and try something different.

9

Person 9

Sue is a student. She will go to university next year, and her parents want to give her a holiday as a present before she goes. Sue likes reading and dancing, and wants somewhere romantic and relaxing.

10

Person 10

Mrs Wallis wants to give her husband a holiday to celebrate his 60th birthday. Mr Wallis likes all sorts of sport, but because his heart is rather weak, he cannot do anything too energetic. Mrs Wallis would like to stay in a good hotel.

11

Person 11

Terry owns his own business. He has had a successful year, and he wants a holiday. He wants somewhere with very good food and accommodation, and he wants to enjoy himself without doing very much.

12

Person 12

The Hamble family are looking for something that will interest Mark aged 7 and Tammy, aged 13. They all like animals and sport. Mr and Mrs Hamble would also like something that lets them do some exploring on their own.

A) Golf in America: A golfing safari across the United States in just ten days. Try eight great courses from New England to California, with golf lessons as you go. Transport is provided between your 5 star hotels and golf courses.

B) Climbing in Switzerland: Whether you are experienced or just beginning, Hipeak have the mountaineering holiday for you. Choose one of our seven, ten or fourteen day packages, and you can enjoy wonderful alpine scenery while climbing at your particular level of ability. No mountaineering gear is needed – we supply it all.

C) Florida and Disney World: Every kid dreams of Disney World, and after mom and dad have seen it all, you can leave the children with one of our special guides, and try some scuba diving in the Florida Keys, and a shopping tour of Miami. Or you can see Florida for yourselves with our no-problem car-hire scheme.

D) Safari in Kenya: Try seven days in Wariwaro lodge in Kenya's Serengetti park. Coach tours through the savannah to see some of the world's most spectacular wildlife. Flights go every Sunday from Heathrow, London, and there is our special family option, with children getting a one-third discount.

E) Holiday in Majorca: Two wild weeks in Majorca. Yes, if you are aged between 18 and 31, this package holiday is for you. The fun starts at Manchester airport, and in Majorca the party moves from the beach to the disco to the swimming pool. It only stops when you can't party any more. If you want a challenge, try this!

F) Mediterranean cruise: The ancient Mediterranean in 16 days. Take a luxury cruise on the RMV Perikles to Herculaneum, Caesarea, Athens and Alexandria. Enjoy our on-board facilities, including restaurant, library, cinema and disco.

G) Caribbean Beach: You've been working too hard! Forget it all on a tropical beach. Ten days in a luxury hotel, right on the beach. You hardly have to move from your deckchair, except to enjoy our award-winning cookery. Beach cricket and barbeques are on offer for the more energetic.

H) Hiking in Katmandu: Try a Katmandu adventure. See the Himalayas for yourself in this two week hiking tour. You must be fit and an experienced hiker, as well as having your own boots. We supply tents, the guides and the holiday of a lifetime!

PART 3

Questions 13–19

Read the text and choose the correct heading for each paragraph from the list of headings below. There are more headings than paragraphs, so you will not use all of them. You cannot use any heading more than once.

List of Headings:

I) Just in time
II) Just in case
III) Eventful life
IV) A curious case
V) Reduced Expectations
VI) Royal brother
VII) Royal ancestor
VIII) Double trouble

13. Paragraph A **14. Paragraph B** **15. Paragraph C**
16. Paragraph D **17. Paragraph E** **18. Paragraph F**
19. Paragraph G

A. I am a mother of identical, mirror-image boys — David and John. No one but me can tell them apart. I am constantly amazed at how close they are. Once when they were babies David was ill, but it was John who began crying wildly. I tried to calm John first since nothing was wrong with him. But he only cried louder. Finally I gave some medicine to David — who really was unwell. As soon as John sensed his brother felt better, he immediately settled to sleep.

B. The 12 year old was playing near the Platte River in North Bend, Nebraska. The river was high and as the boy stepped in, the current pushed his legs away. He floated off, spinning in the powerful current. At the last possible moment before the rapids, his yells were heard by his dog. It jumped in, reached the boy and towed him ashore. Another second and the boy would have been swept away to certain death.

C. Armgaard Karl Graves, referred to in press reports as 'the Glasgow Spy', was convicted in Scotland under the Official Secrets Act (1911) for spying on the British Navy. He spent years successfully creating an identity as an Australian doctor and in Scotland even conducted important clinical experiments. But he was eventually caught by a suspicious post office worker as he sent and received post under a variety of assumed names.

D. Zsa Zsa Gabor was born in Budapest on February 6th, 1917. Now in her 90s she has had a long and varied life. She was a beauty queen and singer before becoming a famous screen actress. She was married 8 times but only had one child with second husband, Conrad Hilton. Her last marriage to Frederic von Anhalt gave her the honorary title Prinzessin von Anhalt.

E. "Who do you think you are" is one of my favourite TV programs. Each episode researches the family history of a celebrity, back into the mists of time. In the UK there are good records of births, marriages and deaths going back hundreds of years. One of the best episodes was on Boris Johnson, the Mayor of London. He was thrilled when he discovered he was directly related to King George II.

F. Paris Hilton is a famous socialite, media personality, actress, model and singer. In 2007 her grandfather Barron Hilton pledged 97% of his estate — a value of more than 2 billion US dollars — to a charitable foundation. Many now believe that Paris and the other grandchildren have had their potential inheritance sharply reduced. Others have commented that this news was unlikely to change her future life style.

G. Andy always travels well equipped for any potential possibility. He has a sewing repair kit and a small medical kit with aspirin. These are, I suppose, perfectly sensible. But what about a ball of string, tape measure, masking tape, Swiss army penknife, disposable cutlery, disinfectant, dry bags and an inflatable back rest? Andy says you never know what might happen and it's always best to be prepared.

PART 4

Questions 20–28

Read the following text for questions 20–28.

HOW BACTERIA INVENTED GENE EDITING

1. This week the UK Human Fertilisation and Embryology Authority okayed a proposal to modify human embryos through gene editing. The research, which will be carried out at the Francis Crick Institute in London, should improve our understanding of human development. It will also undoubtedly attract controversy – particularly with claims that manipulating embryonic genes is a first step towards designer babies. Those concerns shouldn't be ignored. After all, gene editing of the kind that will soon be undertaken at the Francis Crick Institute doesn't occur naturally in humans or other animals.

2. It is, however, a lot more common in nature than you might think, and it's been going on for a surprisingly long time – revelations that have challenged what biologists thought they knew about the way evolution works. We're talking here about one particular gene editing technique called CRISPR-Cas, or just CRISPR. It's relatively fast, cheap and easy to edit genes with CRISPR – factors that explain why the technique has exploded in popularity in the last few years. But CRISPR wasn't dreamed up from scratch in a laboratory. This gene editing tool actually evolved in single-celled microbes.

3. CRISPR went unnoticed by biologists for decades. It was only at the tail end of the 1980s that researchers studying Escherichia coli noticed that there were some odd repetitive sequences at the end of one of the bacterial genes. Later, these sequences

would be named Clustered Regularly Interspaced Short Palindromic Repeats – CRISPRs. For several years the significance of these CRISPRs was a mystery, even when researchers noticed that they were always separated from one another by equally odd 'spacer' gene sequences.

4. Then, a little over a decade ago, scientists made an important discovery. Those 'spacer' sequences look odd because they aren't bacterial in origin. Many are actually snippets of DNA from viruses that are known to attack bacteria. In 2005, three research groups independently reached the same conclusion: CRISPR and its associated genetic sequences were acting as a bacterial immune system. In simple terms, this is how it works. A bacterial cell generates special proteins from genes associated with the CRISPR repeats (these are called CRISPR associated – Cas – proteins). If a virus invades the cell, these Cas proteins bind to the viral DNA and help cut out a chunk. Then, that chunk of viral DNA gets carried back to the bacterial cell's genome where it is inserted – becoming a spacer. From now on, the bacterial cell can use the spacer to recognise that particular virus and attack it more effectively.

5. These findings were a revelation. Geneticists quickly realised that the CRISPR system effectively involves microbes deliberately editing their own genomes – suggesting the system could form the basis of a brand new type of genetic engineering technology. They worked out the mechanics of the CRISPR system and got it working in their lab experiments. It was a breakthrough that paved the way for this week's announcement by the HFEA. Exactly who took the key steps to turn CRISPR into a useful genetic tool is, however, the subject of a huge controversy. Perhaps that's inevitable – credit for developing CRISPR gene editing will probably guarantee both scientific fame and financial wealth.

6. Beyond these very important practical applications, though, there's another CRISPR story. It's the account of how the discovery of CRISPR has influenced evolutionary biology. Sometimes overlooked is the fact that it wasn't just geneticists who were excited by CRISPR's discovery – so too were biologists. They realised CRISPR was evidence of a completely unexpected parallel between the way humans and bacteria fight infections. We've known for a long time that part of our immune system *"learns"* about the pathogens it has seen before so it can adapt and fight infections better in future. Vertebrate animals were thought to be the only organisms with such a sophisticated adaptive immune system. In light of the discovery of CRISPR, it seemed some bacteria had their own version. In fact, it turned out that lots of bacteria have their own version. At the last count, the CRISPR adaptive immune system was estimated to be present in about 40% of bacteria. Among the other major group of single-celled microbes – the archaea – CRISPR is even more common. It's seen in about 90% of them. If it's that common today, CRISPR must have a history stretching back over millions – possibly even billions – of years. "It's clearly been around for a while," says Darren Griffin at the University of Kent.

7. The animal adaptive immune system, then, isn't nearly as unique as we thought. And there's one feature of CRISPR that makes it arguably even better than our adaptive immune system: CRISPR is heritable. When we are infected by a pathogen, our adaptive immune system learns from the experience, making our next encounter with that pathogen less of an ordeal. This is why vaccination is so effective: it involves priming us with a weakened version of a pathogen to train our adaptive immune system. Your children, though, won't benefit from the wealth of experience locked away in your adaptive immune system. They have to experience an infection – or be vaccinated – first hand before they can learn to deal with a given pathogen.

Do the following statements agree with the given information? Write TRUE if the statement agrees with the information, FALSE if the statement contradicts the information, NOT GIVEN if there is no information on this.

20 **The research carried out at the Francis Crick Institute in London is likely to be controversial.**
A) True
B) False
C) Not Given

21 **Gene editing, like the one in the upcoming research, can happen naturally in humans or other animals.**
A) True
B) False
C) Not Given

22 **CRISPR-Cas is a gene editing technique.**
A) True
B) False
C) Not Given

23 **CRISPR was noticed when the researchers saw some odd repetitive sequences at the ends of all bacterial genes.**
A) True
B) False
C) Not Given

24 **A group of American researchers made an important revelation about the CRISPR.**
A) True
B) False
C) Not Given

Choose the correct letter, A, B, C or D. Write the correct letter in boxes 25–28 on your answer sheet.

25 **'Spacer' sequences look odd because:**
A) they are a bacterial immune system
B) they are DNA from viruses
C) they aren't bacterial in origin
D) all of the above

26 **The ones, who were excited about the CRISPR's discovery, were:**
A) biologists
B) geneticists
C) physicists
D) A and B

27 **Word "*learns*" in the 6th paragraph means:**
A) determines
B) gains awareness
C) adapts
D) studies

28 **What makes CRISPR better than even our adaptive immune system?**

A) long history of existence
B) immortality
C) heritability
D) adaptiveness

PART 5

Questions 29–35

Read the text below and answer Questions 29–35.

PENGUINS' ANTI-ICE TRICK REVEALED

"What we learn here is how penguins combine oil and nano-structures on the feathers to produce this effect to perfection," explains Kavehpour. By analysing feathers from different penguin species, the researchers discovered Antarctic species the gentoo penguin (Pygoscelis papua) was more superhydrophobic compared with a species found in warmer climes – the Magellanic penguin (Spheniscus magellanicus) – whose breeding sites include Argentinian desert.

Gentoo penguins' feathers contained tiny pores which trapped air, making the surface hydrophobic. And they were smothered with a special preening oil, produced by a gland near the base of the tail, with which the birds cover themselves. Together, these properties mean that in the wild, droplets of water on Antarctic penguins' superhydrophobic feathers bead up on the surface like spheres – formations that, according to the team, could provide geometry that delays ice formation, since heat cannot easily flow out of the water if the droplet only has minimal contact with the surface of the feather.

"The shape of the droplet on the surface dictates the delay in freezing," explains Kavehpour. The water droplets roll off the penguin's feathers before they have time to freeze, the researchers propose. Penguins living in the Antarctic are highly evolved to cope with harsh conditions: their short outer feathers overlap to make a thick protective layer over fluffier feathers which keep them warm. Under their skin, a thick layer of fat keeps them insulated. The flightless birds spend a lot of time in the sea and are extremely agile and graceful swimmers, appearing much more awkward on land.

Kavehpour was inspired to study Antarctic penguins' feathers after watching the birds in a nature documentary: "I saw these birds moving in and out of water, splashing everywhere. Yet there is no single drop of frozen ice sticking to them," he tells BBC Earth. His team now hopes its work could aid design of better man-made surfaces which minimise frost formation.

"I would love to see biomimicking of these surfaces for important applications, for example, de-icing of aircrafts," says Kavehpour. Currently, airlines spend a lot of time and money using chemical de-icers on aeroplanes, as ice can alter the vehicles' aerodynamic properties and can even cause them to crash.

Complete the sentences below. Write ONLY ONE WORD from the passage for each answer.

QUESTIONS 29–35: SENTENCE COMPLETION

Formations like **29** _____ could provide geometry that delays ice formation.

The delay in freezing is dictated by the **30** _____ of the droplet.

Penguins in Antarctic are highly evolved to be able to cope with **31** _____ conditions.

Penguins are insulated by a **32** _____ layer of fat.

On the land, penguins appear much more **33** _____ than in the sea.

The inspiration came to Kavehpour after watching a **34** _____ about penguins.

Kavehpour would like to see **35** _____ surfaces which minimise frost formation.

MOCK TEST 20

35 answers

1	elephants	13	VIII	25	C
2	clothing	14	I	26	D
3	researchers	15	IV	27	B
4	Kamba	16	III	28	C
5	reacted	17	VII	29	spheres
6	colour	18	V	30	shape
7	researchers	19	II	31	harsh
8	B	20	True	32	thick
9	F	21	False	33	awkward
10	A	22	True	34	documentary
11	G	23	False	35	biomimicking
12	C	24	Not Given		